Exercise: Uncertainty and Variability

This exercise uses:

- the `{ggplot2}` library
- a custom function and applying that

- Your knowledge from past in-class exercises, videos, homework, etc. and corresponding modules from the course site

Overview

This exercise provides some practice creating visualizations that allow you to communicate uncertainty or variability in data. Whereas `geom_bar()` or `geom_col()` *may* be useful for communicating summary statistics like the mean or median, they fail to communicate information about uncertainty or variability in data. By contrast, `geom_histogram()`, `geom_density()`, and variants thereof visualize the full distribution of data but full visual representation of the distribution may not always be needed. In many instances, you need to communicate estimated dispersion statistics, confidence intervals, inter-quartile ranges, or other information about a distribution.

When building statistical models, parameters are estimated but they are not precise and will change based on the size of a data set, the training sample, and other factors. For this reason, confidence intervals are used to provide a range of values within which one may expect to find the true estimated parameter. Smaller samples of data will often have larger confidence intervals simply because fewer data points are used to estimate the interval. Thus, point estimates use to compare athletes, salaries, days unemployed, energy draw, days on vacation, etc. will differ in uncertainty (interval estimate) based on inherent variability and the number of data points.

Whether you are representing data or building statistical models, understanding and representing variability is an important aspect of data visualization. For this reason, your project *must* include some data visualization that captures an essence of uncertainty and this exercise is designed to help you by providing a way for you to represent confidence in estimated data for your project.

A research paper on misinterpretation of error bars by *Krzywinski & Altman (2013)* can be found **here** if you are uncertain about comparing distributions by their standard errors and/or confidence intervals.

Data Set

Use the `SWIM` (accessible here or your project data.

Custom Function

The following function will take a vector, compute its mean and standard error in order to estimate the confidence interval. The function returns a data frame with the following three column variables: `y`, `ymin`, and `ymax`. These three variables represent the *mean* as well as the *upper* and *lower* limits of the confidence interval. Use the code to create a function in your code so that you can use it in your answer.

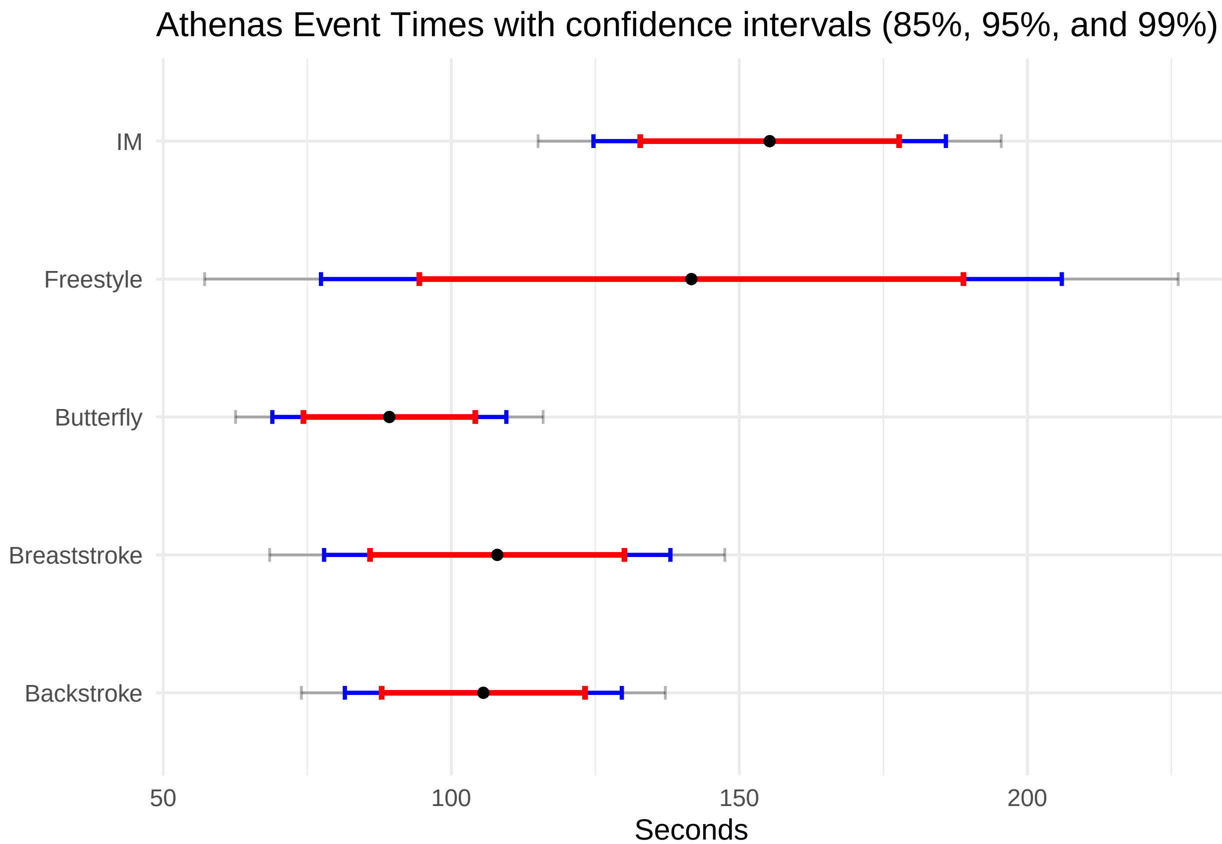
```
mean_ci <- function(x, level = 0.95) {
  x = na.omit(x)
  m = mean(x)
  se = sd(x) / sqrt(length(x))
  ci =  qnorm(1 - (1 - level) / 2) * se

  return(
    data.frame(y = m, ymin = m - ci, ymax = m + ci)
  )
}
```

NOTE: Your **custom function** will iterate your grouping variable. As covered in other content, custom function will require the use of the `~` to perform a lambda function on a variable represented as `.x` in the function.

Example `~ my_function(.x)`

Problem: Plot Replication

Replicate this Plot as best as you are able.


Athenas Event Times with confidence intervals (85%, 95%, and 99%)
IM
Freestyle
Butterfly
Breaststroke
Backstroke
50
100
150
200
Seconds